

Engineering Chemistry (NCY 101/102)

Comprehensive Unit-Wise Previous Year Questions Compilation

Unit 1: Bonding & Spectroscopy

- What is the AXE method? Illustrate and explain the molecular geometry for different steric numbers with suitable examples.
- Describe the basic concepts of Valence Bond Theory (VBT) and explain hybridization with a suitable example.
- Draw the schematic diagram of an FTIR spectrophotometer and derive the expression used to determine the number of vibrational modes of a molecule under infrared radiation.
- Describe Crystal Field Theory (CFT)? Explain the splitting of d-orbitals in an octahedral field.
- Calculate the CFSE value of d^4 system in Octahedral (strong and weak field ligand) and tetrahedral geometry.
- Write the Beer-Lambert Law of UV-vis spectroscopy. The solution of compound having concentration 0.0001g/l gave 0.2 absorbance value when measuring using 1.0cm cell. Calculate the molar extinction coefficient.
- How would you identify the following pairs of compounds with the help of IR technique? (i) Ethanol and propanone (ii) Methanol and ethanoic acid (iii) Ethylamine and acetamide.
- Write the postulates of Crystal field Theory for octahedral and tetrahedral complex. Show the splitting of d-orbitals of $[\text{Co}(\text{NH}_3)_6]^{3+}$ and fill the metal electrons.
- What is mesophase? Classify the liquid crystal and explain the types of organic molecules working as liquid crystal.
- Draw the Jablonski diagram and explain the terms: Absorbance, fluorescence, phosphorescence, intersystem crossing and internal conversion.
- What is Band Theory? Explain the band gap of conductor, insulator and semiconductor on the basis of band theory of solids.
- Explain the d-d transition in octahedral complex and tetrahedral complex considering different types of spin. Calculate the CFSE value for the complex $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$.
- What is chromophore and auxochrome in UV-Vis Spectroscopy? What types of shifts occur in the spectra of UV-Vis spectroscopy and what do they signify?
- What is the working principle of UV-Vis spectroscopy and how can the spectrum be interpreted to identify different types of bonds in a sample?